

Chapter Two: Computer System

2. Introduction to computer systems

A computer system is a **set of integrated devices that input, output, process, and store data and information**. Computer systems are currently built around at least one digital processing device. There are four main hardware components in a computer system: Input, Storage, Output and Communication devices (CPU).

A **computer system** is a basic, complete and functional hardware and software setup with everything needed to implement computing performance.

It is a collection of entities (hardware, software) that are designed to receive, process, manage and present information in a meaningful format.

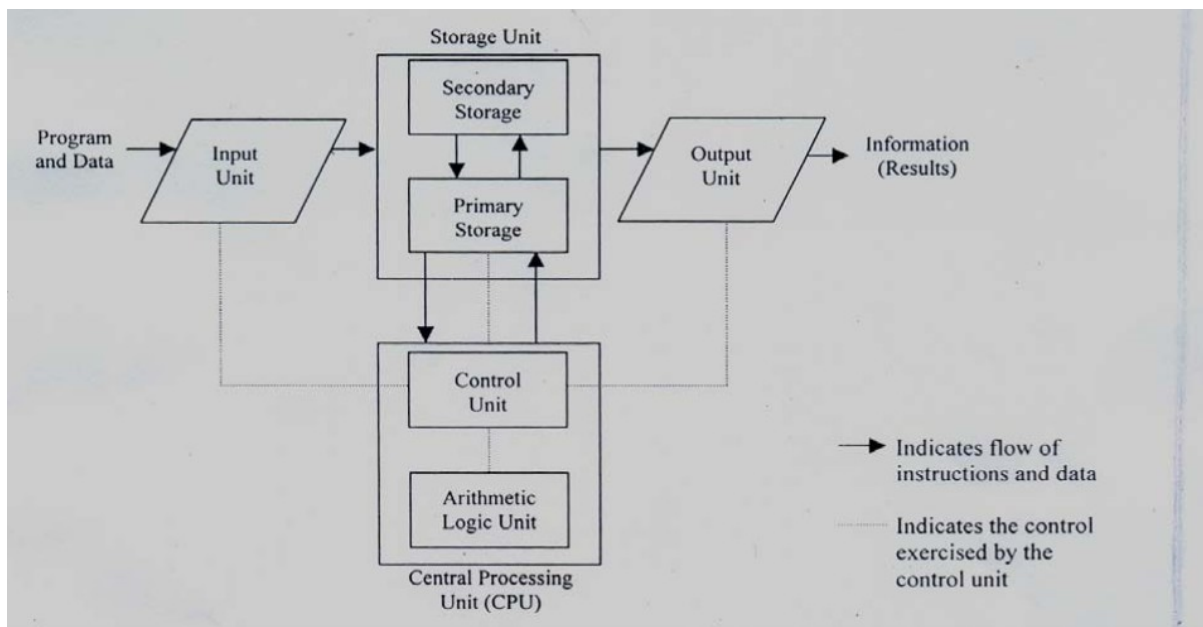


Figure2.1 Basic organization of a computer system

Generally, a computer system is composed of two main components:

1. Computer hardware and
2. Computer software

2.1. Computer Hardware

Computer hardware is the physical part of the computer system that can be seen and felt. The hardware part of a computer system is composed of a number of interacted physical parts. Hardware is an important part of our computer system without which the computer is incomplete.

There are various **hardware components of computer** system, which are given below.

♣ Mouse

♣ Keyboard

♣ Monitor

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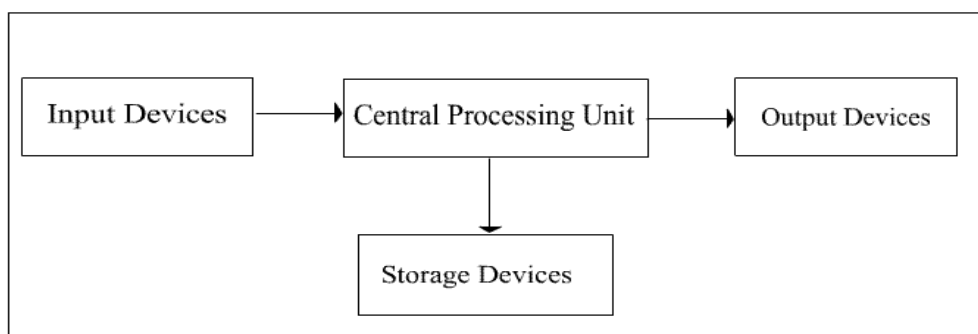
♣ Motherboard	♣ Optical Drive	♣ Sound Card
♣ CPU	♣ IO System	♣ Video Display
♣ RAM Memory	♣ Power Supply	♣ Controller
♣ ROM Memory	♣ External Ports	
♣ Hard Disk Drive	♣ CPU Fan	

1.1 Types of Computer Hardware

The hardware part of a computer system is composed of a number of interacting physical parts based on the need of the information flow. Information flows in the computer hardware. There are several criteria by which computer hardware can be categorized.

Based on **information processing**, we can divide **computer hardware** into **four**:

- i) Input Devices
- ii) Storage Devices
- iii) Output Devices
- iv) Central Processing Unit (CPU)



The operations of computer components are given below:

- 1) Inputting:** It is the process of entering raw data, instructions and information into the computer. It is performed with the help of input devices.
- 2) Storing:** The computer has primary memory and secondary storage to store data and instructions. It stores the data before sending it to CPU for processing and also stores the processed data before displaying it as output.
- 3) Processing:** It is the process of converting the raw data into useful information. This process is performed by the CPU of the computer. It takes the raw data from storage, processes it and then sends back the processed data to storage.
- 4) Outputting:** It is the process of presenting the processed data through output devices like monitor, printer and speakers.

i) Input Devices

Input Devices are the electronic parts of the Computer. The **input devices of Computers** are used to input the user's data or information into computers for the output results.

Input Devices can be defined as an electro-mechanical device that allows a user to input the data into a computer, by usually typing on a keyboard, or by clicking a mouse, we input data into the Computer.

The data is processed into the main memory through the input device. The Computer accepts instructions from the user and converts the received instruction into machine language. **Input devices** are used to **enter information** into **computer**. They **convert** the **data** we give them into the form that can be **manipulated** in the **computer** (**electronic format**).

Input device enables the user to send data, information, or control signals to a computer. The Central Processing Unit (CPU) of a computer receives the input and processes it to produce the output.

Some of the popular input devices are:

- | | | |
|---------------|-----------------------|-------------------------|
| 1. Keyboard | 8. Magnetic Ink | 13. Gesture recognition |
| 2. Mouse | Character Recognition | devices |
| 3. Scanner | (MICR) | 14. Light Gun |
| 4. Joystick | 9. Optical Character | 15. Touch Pad |
| 5. Light Pen | Reader (OCR) | 16. Remote |
| 6. Digitizer | 10. Digital Camera | 17. Touch screen |
| 7. Microphone | 11. Paddle | 18. VR |
| | 12. Steering Wheel | 19. Webcam |
| | | 20. Biometric Devices |

1. Keyboard



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The keyboard is a basic input device that is used to enter data into a computer or any other electronic device by pressing keys. It has different sets of **keys** for **letters, numbers, characters, and functions**. Keyboards are connected to a computer through USB or a Bluetooth device for wireless communication.

2) Mouse



The mouse is a hand-held input device which is used to move cursor or pointer across the screen. It is designed to be used on a flat surface and generally has left and right button and a scroll wheel between them. Laptop computers come with a touchpad that works as a mouse. It lets you control the movement of cursor or pointer by moving your finger over the touchpad. Some mouse comes with integrated features such as extra buttons to perform different buttons.

3) Scanner



The scanner uses the pictures and pages of text as input. It scans the picture or a document. The scanned picture or document then converted into a digital format or file and is displayed on the screen as an output. It uses optical character recognition techniques to convert images into digital ones.

4) Digitizer



Digitizer is a computer input device that has a flat surface and usually comes with a stylus. It enables the user to draw images and graphics using the stylus as we draw on paper with a pencil. The images or graphics drawn on the digitizer appear on the computer monitor or display screen. The software converts the touch inputs into lines and can also convert handwritten text to typewritten words.

It can be used to capture handwritten signatures and data or images from taped papers. Furthermore, it is also used to receive information in the form of drawings and send output to a CAD (Computer-aided design) application and software like [AutoCAD](#). Thus, it allows you to convert hand-drawn images into a format suitable for computer processing.

5) Microphone



The microphone is a computer input device that is used to input the sound. It receives the sound vibrations and converts them into audio signals or sends to a recording medium. The audio signals are converted into digital data and stored in the computer. The microphone also enables the user to telecommunicate with others. It is also used to add sound to presentations and with webcams for video conferencing.

6) Magnetic Ink Character Recognition (MICR)



MICR computer input device is designed to read the text printed with magnetic ink. MICR is a character recognition technology that makes use of special magnetized ink which is sensitive to magnetic fields. It is widely used in banks to process the cheques and other organizations where security is a major concern. It can process three hundred cheques in a minute with hundred-percent accuracy. The details on the bottom of the cheque (MICR No.) are written with magnetic ink. A laser printer with MICR toner can be used to print the magnetic ink. The device reads the details and sends to a computer for processing. A document printed in magnetic ink is required to pass through a machine which magnetizes the ink, and the magnetic information is then translated into characters.

7) Optical Character Reader (OCR)



OCR computer input device is designed to convert the scanned images of handwritten, typed or printed text into digital text. It is widely used in offices and libraries to convert documents and books into electronic files.

It processes and copies the physical form of a document using a scanner. After copying the documents, the OCR software converts the documents into a two-color (black and white), version called bitmap. Then it is analyzed for light and dark areas, where the dark areas are selected as characters, and the light area is identified as background. It is widely used to convert

hard copy legal or historic documents into PDFs. The converted documents can be edited if required like we edit documents created in ms word.

Digital camera:



It is a digital device as it captures images and records videos digitally and then stores them on a memory card. It is provided with an image sensor chip to capture images, as opposed to film used by traditional cameras. Besides this, a camera that is connected to your computer can also be called a digital camera.

It has photosensors to record light that enters the camera through the lens. When the light strikes the photosensors, each of the sensors returns the electrical current, which is used to create the images.

Remote:



It is a [hardware device](#) designed to control the functioning of a device, e.g., a TV remote that can be used to change channels, increase or decrease the volume, from a distance without leaving the seat. The first cordless TV remote was invented by Dr. Robert Adler of Zenith in 1956. The remote sends the electromagnetic waves to communicate with the device. These waves can be infrared rays, radio waves, etc.

Touch screen:



It is the display screen of a device such as a smartphone, tablet, etc., that allows users to interact or provide inputs to the device by using their finger. Today, most of the electronic devices come with touchscreen as an alternative to a mouse for navigating a graphical user interface. For example, by touching, you can unlock your phone, open emails, open files, play videos, etc. Besides this, it is used in lots of devices such as Camera, Car [GPS](#), Fitness machine, etc. The concept of the touch screen was first introduced and published by E.A. Johnson in 1965. The first touch screen was developed at the beginning of the 1970s by CERN engineers Frank Beck and Bent Stumpe.

Biometric Devices:

[Biometrics](#) refers to a process in which a person is identified through his or her biological features such as fingerprints, eye cornea, face structure, etc. It is done by using biometric devices, which can be of different types based on their scanning features and abilities, such as:

i) Face Scanner:



It is designed to identify a person by scanning his or her face. It takes the face measurements of a person. For example, the distance between eyes, nose, and mouth, etc., accordingly, it confirms the identity of a person. Besides this, it is smart enough to differentiate between a person's picture and the real person.

ii) Hand Scanner:



The hand of a person can also be used to verify his or her identity as every person has a unique pattern of veins in the palm, just like fingerprints. This device takes advantage of this feature; it identifies a person by scanning the palm of his hand. It uses infrared light to scan veins' patterns and blood flowing in them. Palm is even more unique than fingerprints.

iii) Fingerprint Scanner:

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It scans the fingerprints to identify people or for biometric authentication. This device is developed, keeping in mind the fact that no two persons in the world can have the same fingerprints. It is widely used in companies as a fingerprint attendance system to mark the attendance of employees. This type of scanners captures the pattern of valleys and ridges found on a finger and store it in the memory or database. When you press your finger on the given space, it verifies the identity by using its pattern-matching software.

iv) Retina or Iris Scanner:



It scans the retina or iris of a person's eye to confirm the identity. This device is more secure than others as it is next to impossible to copy the retina or iris. It works by mapping the retina's blood vessel patterns of the eye. The blood vessels of retina absorb light more easily as well as can be identified with appropriate lighting.

In this scan, a beam of low-energy infrared light falls on the retina through the scanner's eyepiece. Then, the software captures the network of blood vessels in the retina and uses it to verify a person's identity.

v) Voice Scanner:



It records the voice of a person and digitizes it to create a distinctive voice print or template. The voiceprints are stored in the database, and are used to verify the voice of a person to confirm his or her identity. The person is required to speak in the normal or same voice that was used to create a voice template. It is not much reliable as it can be misused using a tape recording.

Output Devices

The output device displays the result of the processing of raw data that is entered in the computer through an input device. There are a number of output devices that display output in different ways such as text, images, hard copies, and audio or video.

Some of the popular output devices are:

1. Monitor
 - CRT Monitor
 - LCD Monitor
 - LED Monitor
 - Plasma Monitor
2. Printer
3. Projector
4. Speaker

1) Monitor

The monitor is the display unit or screen of the computer. It is the main output device that displays the processed data or information as text, images, audio or video.

2) Printer

A printer produces hard copies of the processed data. It enables the user, to print images, text or any other information onto the paper.

3) Projector



A projector is an output device that enables the user to project the output onto a large surface such as a big screen or wall. It can be connected to a computer and similar devices to project their output onto a screen. It uses light and lenses to produce magnified texts, images, and videos. So, it is an ideal output device to give presentations or to teach a large number of people.

Modern projects (digital projectors) come with multiple input sources such as HDMI ports for newer equipment and VGA ports that support older devices. Some projectors are designed to support Wi-Fi and Bluetooth as well. They can be fixed onto the ceiling, placed on a stand, and more and are frequently used for classroom teaching, giving presentations, home cinemas, etc.

A digital projector can be of two types:

Liquid Crystal Display (LCD) digital projector: This type of digital projectors are very popular as they are lightweight and provide crisp output. An LCD projector uses transmissive technology to produce output. It allows the light source, which is a standard lamp, to pass through the three colored liquid crystal light panels. Some colors pass through the panels and some are blocked by the panels and thus images are on the screen.

Digital Light Processing (DLP) digital projector: It has a set of tiny mirrors, a separate mirror for each pixel of the image and thus provide high-quality images. These projectors are mostly used in theatres as they fulfill the requirement of high-quality video output.

Central Processing Unit (CPU)

A Central Processing Unit is also called a processor, central processor, or microprocessor. It carries out all the important functions of a computer. It receives instructions from both the hardware and active software and produces output accordingly. It stores all important programs like operating systems and application software. CPU also helps Input and output devices to communicate with each other. Owing to these features of CPU, it is often referred to as the brain of the computer.

CPU is installed or inserted into a CPU socket located on the motherboard. Furthermore, it is provided with a heat sink to absorb and dissipate heat to keep the CPU cool and functioning smoothly.

CPU executes instructions and performs the computer's processing activities. It is also known as processor or microprocessor. It functions the same purpose as the human brain for human being. It is called the brain of the computer.

One of the basic features of a computer that affects its entire performance is the CPU speed. CPU speed is measured in Hertz (Hz). **Hertz** is the number of cycles per second.

1Hz=1cycle per second. Larger units are KHz (Kilo Hertz), MHz (Mega Hertz), GHz (Giga Hertz), etc.

$$1 \text{ KHz} = 1000 \text{ Hz}$$

$$1 \text{ MHz} = 1000 \text{ KHz}$$

$$1 \text{ GHz} = 1000\text{MHz}$$

Current CPUs are as fast as 2-3GHz (2-3 billion cycles per second)

CPU has two sub-components:

- ✓ Control Unit (CU)
- ✓ Arithmetic Logic Unit (ALU)

Control Unit

As human brain controls the body, control unit controls the computer hardware. Control Unit does not execute instruction by itself, i.e. does not carry out instruction processing, but it directs other processing elements to execute instructions.

It makes use of electrical signals to instruct the computer system for executing already stored instructions. It takes instructions from memory and then decodes and executes these instructions. So, it controls and coordinates the functioning of all parts of the computer. The

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Control Unit's main task is to maintain and regulate the flow of information across the processor. It does not take part in processing and storing data.

Arithmetic Logic Unit (ALU)

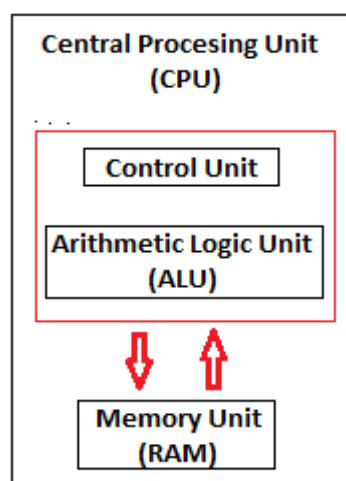
ALU: It is the arithmetic logic unit, which performs arithmetic and logical functions. Arithmetic functions include addition, subtraction, multiplication division, and comparisons. Logical functions mainly include selecting, comparing, and merging the data. A CPU may contain more than one ALU. Furthermore, ALUs can be used for maintaining timers that help run the computer.

The purpose of ALU is to execute instruction. It performs two operations:

- ✓ Arithmetic operation
- ✓ Logic Operation

Arithmetic operation: this includes mathematical operations like addition, subtraction, multiplication, division, etc. If you give your computer the instruction $2+3$, this will be included in arithmetic operation and it is executed by Arithmetic Unit.

Logical Operation: this is concerned with the comparison of data and it is called logical operation. It includes operators like less than, greater than, equal to, less or equal to, greater or equal to, different from, etc. e.g. if **mark>80**, grade is 'A'.



Storage Devices

One of the unique features of computers is storage. Data can be stored on different storage media temporarily or permanently. Storage devices can be categorized into to as:

1. Primary storage device
2. Secondary storage device

1. Primary Memory / Main Memory

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Primary memory, also called Main memory, refers to integrated circuit that stores program instructions and data. The CPU closely works with the main memory to perform its activities.

Memory stores three things:

- ✓ Operation system software instructions
- ✓ Application software instruction
- ✓ Data that is being processed

Depending on the type of information they store and the technology used, the primary memory can be categorized into three:

- ✓ RAM (Random Access Memory)
- ✓ ROM (Read Only Memory)

RAM is temporary storage i.e. the data is lost when the computer is off unlike secondary storage. Because of this it is called volatile memory.

Why is it volatile? It uses electric power to store data. When you write anything on your computer, first it is stored on RAM. When you save the file, it is transferred into secondary storage.

It is directly accessible by CPU. It is called RAM because each memory location can be accessed randomly using memory address.

RAM is:

- Small in storage capacity
- Fast in processing speed
- Volatile
- Expensive price wise.

ROM stores data and programs that are permanently required by the computer. They have programs built into them at the factory and that program could not be changed or erased by the user, but read. It is non-volatile, read-only (not changeable) memory. Read-only means data can't be altered or erased but read.

ROMs are mainly used to store programs and data which do not change and are frequently used known as micro programs. A good example of micro program stored in a ROM chip is the set of instructions that is needed to make the computer system ready for use when its power is switched on. This micro program called 'system boot program', contains a set of start-up instructions to check if the system hardware like memory, I/O devices, etc. are functioning properly, and looks for an operating system and loads its core part in the volatile RAM of the system to produce the initial display-screen prompt.

2. Secondary (auxiliary) Storage

- ♣ Secondary storage (also called auxiliary storage) supplements the primary memory.

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- ♣ The fundamental characteristics of secondary storage are high capacity and low cost, although speed, reliability and portability might also be important.
 - ♣ It takes many forms.
 - ♣ It includes **punched cards, punched paper tape, magnetic tape, magnetic disk and optical disk, SD cards, flash drives, and Cloud storage.**
 - ♣ **Secondary storages:**
 - Are non-volatile, they store data and information permanently,
 - Are cheap or low in cost,
 - Operating speed far slower than the primary storage.
 - Store large amount of information.
 - ♣ Based on information access, secondary storage devices are divided into two:
- ✓ **Sequential Access:** information is accessed sequentially. To access information on such media, we start from the beginning and read through to the end. Jumping to some part is not possible. E.g. tape recorder cassette
 - ✓ **Random Access:** information can be accessed in any order. You can access the first or the last part of information by jumping others. E.g. floppy disk

2.2. Computer Software

Computer hardware is directed by a set of instructions. Without these instructions, computers can do nothing. These set of instructions are called software (also called programs). One or more programs are termed as **software**. We use programming languages to write these instructions. Examples of programming languages include C, C++, Java, etc. Software acts as **intermediate** between the **user** and **the electronic components of the computer**.

USER
Software
Hardware

Software is categorized into two:

1. System Software
2. Application Software

1. System Software

System software consists of programs that are related to controlling the actual operations of the computer equipment/resource.

There are three types of system software:

- ✓ Operating System

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- ✓ Utility Software
- ✓ Language translators

A. **Operating system** is system software that manages computer hardware, software resources, and provides common services for computer programs.

- ♣ OS is a software that acts as an interface between computer hardware components and the user.
- ♣ Every computer system must have at least one operating system to run other programs.
- ♣ Applications like Browsers, MS Office, Notepad Games, etc., need some environment to run and perform its tasks.
- ♣ The OS helps you to communicate with the computer without knowing how to speak the computer's language.
- ♣ It is not possible for the user to use any computer or mobile device without having an operating system.
- ♣ It organizes resources such as keyboard, mouse, printer, monitor, etc.
- ♣ It also presents GUI (Graphical User Interface) to the user for easy use of computer.
- ♣ It makes complex hardware more users friendly i.e. it acts between the user and hardware.
- ♣ Operating system coordinates the activity between the user and the computer

Types of Operating Systems

Operating systems can be classified by:

- ✓ The number of programs they can handle at a time and
- ✓ The number of users they serve at once at one or different stations (i.e. terminals or microcomputers connected to a central computer).

I. Single tasking operating systems

With single tasking operating systems only one program can be run on a computer at a time. In order to run another program, one must remove the first program loaded in the computers main memory and load the other one (i.e. it can't handle two or more programs at a time) These types of operating systems are single user or can serve only one user at a time.

II. Multi user operating system

- ✓ It supports a number of work stations connected to a central system.
- ✓ A number of users can use the resources of one high capacity computer by the help of terminals.

III. Real Time Operating System

- ✓ A real time is a system that is capable of processing data so quickly that the results are available to influence the activity currently taking place.

Example: - Air plane seat reservations and computer controlled plant.

- Its primary characteristics are that it responds to an event within a well-defined time.

B. Utilities software

- ♣ It is **programs** that make computing easier.
- ♣ They perform specific tasks related to managing computer resources or files.

Eg. Backup software, antivirus software, disk management, file management.

C. Language translators

- ♣ Are used to **convert the programming instruction written by users into binary code** that the computer can understand.
- ♣ They are **written for specific programming languages and computer system**.

2. Application Software

Application software performs **useful work** for the **user**. It is software that is designed to perform tasks for the specific area or areas. But for use in more than one installation.

These useful works could be:

- ✓ Word processing-document creation
- ✓ Spreadsheet-electronic calculation
- ✓ Data base system
- ✓ Email/communicating-email sending and reading
- ✓ Users use this software to perform different activities like calculation, video editing, word processing, presentation, etc.